

Problem 18.5

A perpetuity-immediate pays X per year. Brian receives the first n payments, Colleen receives the next n payments, and Jeff receives the remaining payments. Brian's share of the present value of the original perpetuity is 40%, and Jeff's share is K . Calculate K .

Solution

The present value of the original perpetuity-immediate is

$$PV = \frac{X}{i}.$$

Brian receives the first n payments, so his present value is

$$Xa_{\overline{n}|} = X \left(\frac{1 - v^n}{i} \right).$$

Brian's share of the original present value is 40%, hence

$$X \left(\frac{1 - v^n}{i} \right) = 0.40 \left(\frac{X}{i} \right).$$

Canceling X/i gives

$$1 - v^n = 0.40.$$

Therefore,

$$v^n = 0.60.$$

Jeff receives all payments after the first $2n$ payments. This is the original perpetuity deferred $2n$ years, so Jeff's present value is

$$v^{2n} \left(\frac{X}{i} \right).$$

Thus, Jeff's share of the original perpetuity value is

$$K = v^{2n}.$$

Since $v^n = 0.60$,

$$K = (v^n)^2 = (0.60)^2 = 0.36.$$

Hence,

$$\boxed{K = 36\%}.$$